

Backup and Restore on Same Host Using pg_basebackup

How to Backup and Restore PostgreSQL DB Cluster on the Same Host Using pg_basebackup

Table of Contents

- 1. Environment
 - 2. Verify Existing DB Setup
 - 3. Pre-requisites
 - 4. Take Backup
 - 5. Prepare & Restore
 - 5.1 Stop PostgreSQL and Clean Directories
 - 5.2 Remove all from Data & WAL directories
 - 5.3 Restore Data and WAL
 - A. Copy Backup files to \$PGDATA
 - B. Copy WAL files from \$PGDATA/pg_wal to /pgWal/pgsql17/wal/
 - C. Remove the existing \$PGDATA/pg_wal directory
 - D. Create a symbolic link pointing \$PGDATA/pg_wal to a separate WAL directory
 - E. Set Permissions
 - 6. Start PostgreSQL
 - 7. Final Verification
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1. Environment

ASPECT	ENV
Hostname	lxicbpdsgv01
IP Address	192.168.2.51

ASPECT	ENV
OS	RHEL 9
DB Version	PostgreSQL v17.6
Archive mode	archive_mode=off
pgData	/pgData/pgsql17/data
WAL Directory	/pgWal/pgsql17/wal
Tablespace	pg_default
Databases	DELL, ORCL

2. Verify Existing DB Setup

```
[postgres@lxicbpgdsgv01 ~]$ psql
psql (17.6)
Type "help" for help.

postgres=# \l+

List of databases
   Name | Owner | Encoding | Locale Provider | Collate | Ctype | Locale | ICU
Rules | Access privileges | Size | Tablespace | Description
-----+-----+-----+-----+-----+-----+-----+-----
--+-+-----+-----+-----+-----+-----+-----+-----
 dell   | postgres | UTF8 | libc | en_SG.UTF-8 | en_SG.UTF-8 | |
 | | 7763 kB | pg_default |
 orcl   | postgres | UTF8 | libc | en_SG.UTF-8 | en_SG.UTF-8 | |
 | | 7907 kB | pg_default |
 postgres | postgres | UTF8 | libc | en_SG.UTF-8 | en_SG.UTF-8 | |
 | | 492 MB | pg_default | default administrative connection database
 template0 | postgres | UTF8 | libc | en_SG.UTF-8 | en_SG.UTF-8 | |
 | =c/postgres +| 7545 kB | pg_default | unmodifiable empty database
 | | | | | |
 | postgres=CTc/postgres | |
 template1 | postgres | UTF8 | libc | en_SG.UTF-8 | en_SG.UTF-8 | |
 | =c/postgres +| 7723 kB | pg_default | default template for new databases
 | | | | | |
 | postgres=CTc/postgres | |
(5 rows)
```

```

postgres=# \db
      List of tablespaces
  Name      |  Owner   | Location
-----+-----+-----
 pg_default | postgres |
 pg_global  | postgres |
(2 rows)

postgres=# \c dell
You are now connected to database "dell" as user "postgres".
dell=# select * from test.emp;
 name | designation | project | company
-----+-----+-----+-----
  Sugi | DBA         | Jetstar | iGATE
  Teja | DBA         | RCM     | iGATE
  RAJ  | DBA         | RCM     | iGATE
(3 rows)

dell=#

```

3. Pre-requisites

- [Verify postgresql.conf](#)

```

postgres=# SHOW wal_level;
 wal_level
-----
 replica <----
(1 row)

postgres=#
postgres=# SHOW max_wal_senders;
 max_wal_senders
-----
          10
(1 row)

postgres=# SHOW archive_mode;
 archive_mode
-----
off

```

```
(1 row)
```

```
postgres=#
```

- **Verify pg_hba.conf**

```
# TYPE  DATABASE        USER            ADDRESS           METHOD
# Local connections for replication (for pg_basebackup run locally)
local   replication    all                                trust

# Remote connections for replication (for pg_basebackup run remotely)
#host   replication    repl_user       192.168.2.52/32   scram-sha-256
```

- **Verify user permissions:REPLICATION or SUPERUSER required**

```
postgres=# \du

                                List of roles

Role name |                               Attributes
-----+-----
postgres | Superuser, Create role, Create DB, Replication, Bypass RLS

postgres=#
```

- **Verify free space for backup**

```
[postgres@lxicbpgdsgv01 ~]$ du -sh /pgData/pgsql17/data/
524M    /pgData/pgsql17/data/
[postgres@lxicbpgdsgv01 ~]$
[postgres@lxicbpgdsgv01 ~]$ du -sh /pgWal/pgsql17/wal/
801M    /pgWal/pgsql17/wal/
[postgres@lxicbpgdsgv01 ~]$

[postgres@lxicbpgdsgv01 ~]$ df -h /pgBackup/
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdd1       100G  746M   100G   1% /pgBackup <----
[postgres@lxicbpgdsgv01 ~]$ mkdir -p
/pgBackup/pgsql17/backup/basebackup_10OCT2025
[postgres@lxicbpgdsgv01 ~]$
[postgres@lxicbpgdsgv01 ~]$ mkdir -p /pgBackup/pgsql17/backup/log/
[postgres@lxicbpgdsgv01 ~]$
```

4. Take Backup

```
[postgres@lxicbpgdsgv01 ~]$ ls -ltrh /pgWal/pgsql17/wal | wc -l
53 <----
[postgres@lxicbpgdsgv01 ~]$
```

pg_basebackup does not include all files from the pg_wal directory in the backup. Instead, it selectively includes only the WAL files required to make the base backup consistent at the point in time the backup was taken.

-- If you want to store the base backup and WAL files in separate backup directories.

```
nohup pg_basebackup -U postgres -D
/pgBackup/pgsql17/backup/basebackup_10OCT2025 --
waldir=/pgBackup/pgsql17/backup/wal_backup -Fp -Xs -P -v >
/pgBackup/pgsql17/backup/log/basebackup_10OCT2025.log 2>&1 &
```

The --waldir option in pg_basebackup is supported only when using the plain format (-Fp), not with the tar format (-Ft).

If we want symlinks preserved → use (both -Fp & --waldir use together) -Fp
--waldir=/pgWal/pgsql17/wal

-- OR --

The directory must be empty: /pgBackup/pgsql17/backup/basebackup_10OCT2025

```
[postgres@lxicbpgdsgv01 ~]$ nohup pg_basebackup -U postgres -D
/pgBackup/pgsql17/backup/basebackup_10OCT2025 -Fp -Xs -P -v >
/pgBackup/pgsql17/backup/log/basebackup_10OCT2025.log 2>&1 &
[1] 4973
[postgres@lxicbpgdsgv01 ~]$
```

```
[postgres@lxicbpgdsgv01 ~]$ cat
/pgBackup/pgsql17/backup/log/basebackup_10OCT2025.log
nohup: ignoring input
pg_basebackup: initiating base backup, waiting for checkpoint to complete
pg_basebackup: checkpoint completed
pg_basebackup: write-ahead log start point: 0/85000028 on timeline 1
pg_basebackup: starting background WAL receiver
pg_basebackup: created temporary replication slot "pg_basebackup_4975"
279413/536037 kB (52%), 0/1 tablespace
(...asebackup_10OCT2025/base/5/16533)
536047/536047 kB (100%), 0/1 tablespace
```

```
(...ckup_10OCT2025/global/pg_control)
536047/536047 kB (100%), 1/1 tablespace
pg_basebackup: write-ahead log end point: 0/85000158
pg_basebackup: waiting for background process to finish streaming ...
pg_basebackup: syncing data to disk ...
pg_basebackup: renaming backup_manifest.tmp to backup_manifest
pg_basebackup: base backup completed <-----
[postgres@lxicbpgdsgv01 ~]$
```

5. Prepare & Restore

5.1 Stop PostgreSQL and Clean Directories

```
[root@lxicbpgdsgv01 ~]# systemctl stop postgresql-17.service
[root@lxicbpgdsgv01 ~]#
[root@lxicbpgdsgv01 ~]# ps -ef | grep postgres
root          5057      3151  0 18:57 pts/0    00:00:00 grep --color=auto
postgres
[root@lxicbpgdsgv01 ~]#
```

5.2 Remove all from Data & WAL directories

```
-- Remove all from PGDATA directory

[root@lxicbpgdsgv01 ~]# rm -rf /pgData/pgsql17/data/*
[root@lxicbpgdsgv01 ~]# ls -ltr /pgData/pgsql17/data/
total 0
[root@lxicbpgdsgv01 ~]#

-- Remove all from WAL directory

[root@lxicbpgdsgv01 ~]# rm -rf /pgWal/pgsql17/wal/*
[root@lxicbpgdsgv01 ~]# ls -ltr /pgWal/pgsql17/wal/
total 0
[root@lxicbpgdsgv01 ~]#
```

5.3 Restore Data and WAL

A. Copy Backup files to \$PGDATA

Since we did not use the -Fp format with the --waldir option, all required WAL files will be included in the backup under the default path: PGDATA/pg_wal.

```
[postgres@lxicbpgdsgv01 ~]$ ls -ltr /pgData/pgsql17/data/
total 0
[postgres@lxicbpgdsgv01 ~]$ cd
/pgBackup/pgsql17/backup/basebackup_10OCT2025/
[postgres@lxicbpgdsgv01 basebackup_10OCT2025]$ cp -Rp *
/pgData/pgsql17/data/
[postgres@lxicbpgdsgv01 basebackup_10OCT2025]$
[postgres@lxicbpgdsgv01 basebackup_10OCT2025]$ ls -ltr
/pgData/pgsql17/data/
total 296
-rw-----. 1 postgres postgres 227 Oct 10 18:41 backup_label
drwx-----. 4 postgres postgres 77 Oct 10 18:41 pg_wal
drwx-----. 7 postgres postgres 59 Oct 10 18:41 base
drwx-----. 4 postgres postgres 68 Oct 10 18:41 pg_logical
drwx-----. 2 postgres postgres 6 Oct 10 18:41 pg_dynshmem
drwx-----. 2 postgres postgres 6 Oct 10 18:41 pg_commit_ts
drwx-----. 2 postgres postgres 110 Oct 10 18:41 log
-rw-----. 1 postgres postgres 1169 Oct 10 18:41
postgresql.conf.bkp_10sep2025
-rw-----. 1 postgres postgres 30702 Oct 10 18:41 postgresql.conf.bkp
-rw-----. 1 postgres postgres 88 Oct 10 18:41 postgresql.auto.conf
drwx-----. 2 postgres postgres 18 Oct 10 18:41 pg_xact
-rw-----. 1 postgres postgres 3 Oct 10 18:41 PG_VERSION
drwx-----. 2 postgres postgres 6 Oct 10 18:41 pg_twophase
drwx-----. 2 postgres postgres 6 Oct 10 18:41 pg_tblspc
drwx-----. 2 postgres postgres 6 Oct 10 18:41 pg_subtrans
drwx-----. 2 postgres postgres 6 Oct 10 18:41 pg_stat_tmp
drwx-----. 2 postgres postgres 6 Oct 10 18:41 pg_stat
drwx-----. 2 postgres postgres 6 Oct 10 18:41 pg_snapshots
drwx-----. 2 postgres postgres 6 Oct 10 18:41 pg_serial
drwx-----. 2 postgres postgres 6 Oct 10 18:41 pg_replslot
drwx-----. 2 postgres postgres 6 Oct 10 18:41 pg_notify
drwx-----. 4 postgres postgres 36 Oct 10 18:41 pg_multixact
-rw-----. 1 postgres postgres 2640 Oct 10 18:41 pg_ident.conf
-rw-----. 1 postgres postgres 5600 Oct 10 18:41 pg_hba.conf
-rw-----. 1 postgres postgres 1171 Oct 10 18:41 postgresql.conf
drwx-----. 2 postgres postgres 4096 Oct 10 18:41 global
-rw-----. 1 postgres postgres 30 Oct 10 18:41 current_logfiles
-rw-----. 1 postgres postgres 227736 Oct 10 18:41 backup_manifest
[postgres@lxicbpgdsgv01 basebackup_10OCT2025]$
```

B. Copy WAL files from \$PGDATA/pg_wal to /pgWal/pgsql17/wal/

```
[postgres@lxicbpgdsgv01 ~]$ ls -ltr /pgWal/pgsql17/wal/
total 0
[postgres@lxicbpgdsgv01 ~]$
[postgres@lxicbpgdsgv01 ~]$ cd /pgData/pgsql17/data/pg_wal/
[postgres@lxicbpgdsgv01 pg_wal]$ cp -Rp * /pgWal/pgsql17/wal
[postgres@lxicbpgdsgv01 pg_wal]$
[postgres@lxicbpgdsgv01 pg_wal]$ ls -ltr /pgWal/pgsql17/wal
total 16384
drwx-----. 2 postgres postgres      6 Oct 10 18:41 summaries
drwx-----. 2 postgres postgres      6 Oct 10 18:41 archive_status
-rw-----. 1 postgres postgres 16777216 Oct 10 18:41
0000000010000000000000000085
[postgres@lxicbpgdsgv01 pg_wal]$ cd
[postgres@lxicbpgdsgv01 ~]$
```

C. Remove the existing \$PGDATA/pg_wal directory

```
[postgres@lxicbpgdsgv01 ~]$ rm -rf /pgData/pgsql17/data/pg_wal
[postgres@lxicbpgdsgv01 ~]$
```

D. Create a symbolic link pointing \$PGDATA/pg_wal to a separate WAL directory

```
[postgres@lxicbpgdsgv01 ~]$ ln -s /pgWal/pgsql17/wal
/pgData/pgsql17/data/pg_wal
[postgres@lxicbpgdsgv01 ~]$
[postgres@lxicbpgdsgv01 ~]$ ls -ld /pgData/pgsql17/data/pg_wal
lrwxrwxrwx. 1 postgres postgres 18 Oct 10 19:43 /pgData/pgsql17/data/pg_wal
-> /pgWal/pgsql17/wal
[postgres@lxicbpgdsgv01 ~]$
```

E. Set Permissions

```
[postgres@lxicbpgdsgv01 ~]$ chown -R postgres:postgres /pgData/pgsql17/data
[postgres@lxicbpgdsgv01 ~]$ chmod 700 /pgData/pgsql17/data
```

```
[postgres@lxicbpgdsgv01 ~]$ chown -R postgres:postgres /pgWal/pgsql17/wal
[postgres@lxicbpgdsgv01 ~]$ chmod 700 /pgWal/pgsql17/wal
```

6. Start PostgreSQL

```
[root@lxicbpgdsgv01 ~]# systemctl start postgresql-17.service
[root@lxicbpgdsgv01 ~]# systemctl status postgresql-17.service
• postgresql-17.service - PostgreSQL 17 database server
   Loaded: loaded (/usr/lib/systemd/system/postgresql-17.service;
   enabled; preset: disabled)
   Active: active (running) since Fri 2025-10-10 19:45:21 +08; 5s ago
     Docs: https://www.postgresql.org/docs/17/static/
   Process: 5230 ExecStartPre=/usr/pgsql-17/bin/postgresql-17-check-db-dir
   ${PGDATA} (code=exited, status=0/SUCCESS)
   Main PID: 5235 (postgres)
     Tasks: 7 (limit: 15835)
    Memory: 34.2M
       CPU: 94ms
    CGroup: /system.slice/postgresql-17.service
            └─5235 /usr/pgsql-17/bin/postgres -D /pgData/pgsql17/data/
            └─5236 "postgres: logger "
            └─5237 "postgres: checkpointer "
            └─5238 "postgres: background writer "
            └─5240 "postgres: walwriter "
            └─5241 "postgres: autovacuum launcher "
            └─5242 "postgres: logical replication launcher "
```

Oct 10 19:45:21 lxicbpgdsgv01.rajasekhar.com systemd[1]: Starting PostgreSQL 17 database server...

Oct 10 19:45:21 lxicbpgdsgv01.rajasekhar.com postgres[5235]: 2025-10-10 19:45:21.177 +08 [5235] LOG: redirecting log output to logging collector process

Oct 10 19:45:21 lxicbpgdsgv01.rajasekhar.com postgres[5235]: 2025-10-10 19:45:21.177 +08 [5235] HINT: Future log output will appear in directory "log".

Oct 10 19:45:21 lxicbpgdsgv01.rajasekhar.com systemd[1]: Started PostgreSQL 17 database server.

```
[root@lxicbpgdsgv01 ~]#
```

7. Final Verification

```
[postgres@lxicbpgdsgv01 ~]$ psql
psql (17.6)
```

Type "help" for help.

postgres=# \l+

List of databases

Name	Owner	Encoding	Locale Provider	Collate	Ctype	Locale	ICU Rules	Access privileges	Size	Tablespace	Description
dell	postgres	UTF8	libc	en_SG.UTF-8	en_SG.UTF-8				7609 kB	pg_default	
orcl	postgres	UTF8	libc	en_SG.UTF-8	en_SG.UTF-8				7753 kB	pg_default	
postgres	postgres	UTF8	libc	en_SG.UTF-8	en_SG.UTF-8				492 MB	pg_default	default administrative connection database
template0	postgres	UTF8	libc	en_SG.UTF-8	en_SG.UTF-8				7545 kB	pg_default	unmodifiable empty database
postgres=CTc/postgres											
template1	postgres	UTF8	libc	en_SG.UTF-8	en_SG.UTF-8				7569 kB	pg_default	default template for new databases
postgres=CTc/postgres											

(5 rows)

postgres=# \db

List of tablespaces

Name	Owner	Location
pg_default	postgres	
pg_global	postgres	

(2 rows)

postgres=# \c dell

You are now connected to database "dell" as user "postgres".

dell=# select * from test.emp;

name	designation	project	company
Sugi	DBA	Jetstar	iGATE
Teja	DBA	RCM	iGATE
RAJ	DBA	RCM	iGATE

(3 rows)

dell=#

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Thank you,

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